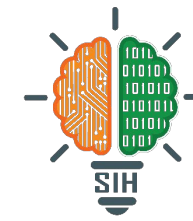


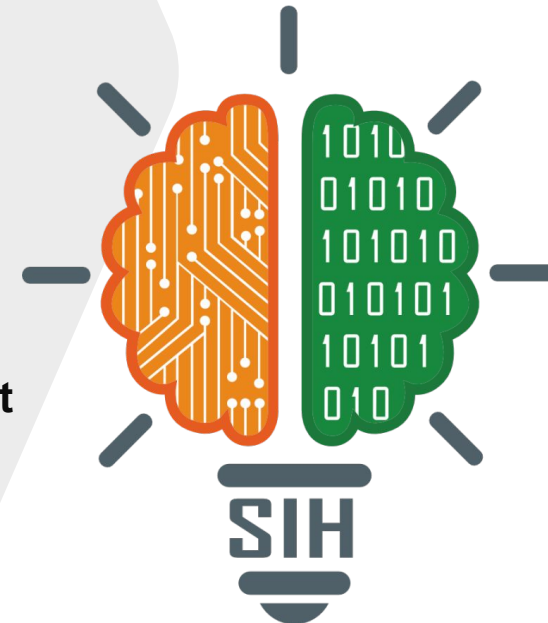
# SMART INDIA HACKATHON 2024



**SMART INDIA  
HACKATHON  
2024**

## TITLE PAGE

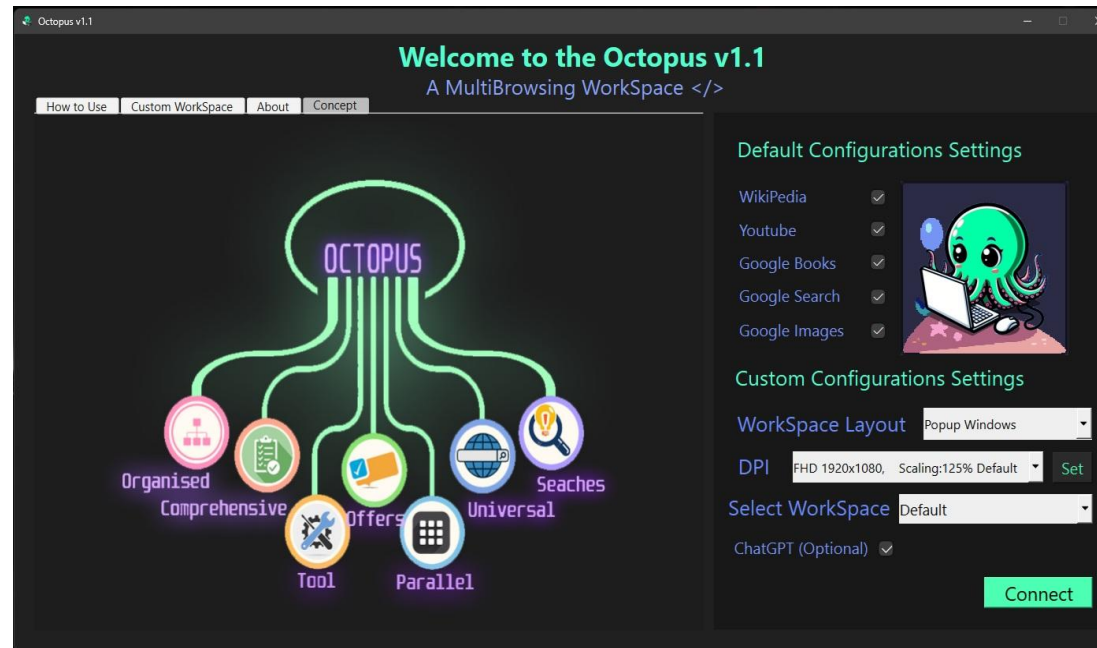
- Problem Statement ID → 1586
- Problem Statement Title → Student Innovation
- Theme → Smart Education
- PS Category → Software
- Organization → AICTE, MIC-Student Innovation
- Team Name → Innovate Addaaa
- Department → Ministry of Education's Innovation Cell (MIC)
- Description → Smart Education, a Concept that Describes learning in digital age. It enables learner to learn more effectively, efficiently, flexibly and comfortably.



# IDEA TITLE

## Project Idea Title & Description

- ❖ We are working on an idea related to (**Student Innovation**)
- ❖ Our initial Project name is (**OCTOPUS v1.1**)
- ❖ Full Form is :- its an **Organised, Comprehensive, Tool, Offers, Parallel & Universal Searches.**
- ❖ In simple words, It's a **Multi Browsing** workspace and Internet based Desktop Application, designed to enhanced the Productivity and efficiency during research and instant information gathering.
- ❖ This tool offers a unique approach to Perform Searching by providing a multi browsing Interface.



## Detailed explanation of Proposed IDEA/Solution

- ❖ It's an AI based Internet Browsing Software or something like that in which when we search or ask anything to the program and within next second it shows all necessary information like **Images, videos, pdf books, Chatgpt results, or multiple website** etc which is sufficient for understanding, in Single screen or in grid like view using Internet and Ai. like in marvel's iron man movie, jarvis shows relevant visualization to the ironman, we are working on this idea, some **AI integration** left, Our project Aims to Simplify the Internet Learning and to increase the **work efficiency** during **productivity** and **Research** related works, and **changes our perspective of Learning from Internet.**

## Innovation and Uniqueness of the solution

- ❖ Other than Commercial Browsers, It has a **centralised Search Bar** located at bottom & central location of the screen.
- ❖ When user search anything in search bar, then it Performs same search to Selected websites simultaneously in which user working on. And it Instantly Pop Ups the results from multiple websites. In front of User.

## Uniqueness of the project

- ❖ **1.User not need to search the same query in each and lots of websites to get the required information.**
- ❖ **2.User Can Search on multiple Websites , using single Search Bar.**

**Instant Notes:-** User can quickly capture important information, just by **Drag & Drop** from multiple websites, in NoteMaker window. & save it to PDF to summarize information, Easily while studying or Browsing content.

# TECHNICAL APPROACH

## Technologies we Use in this Project:-

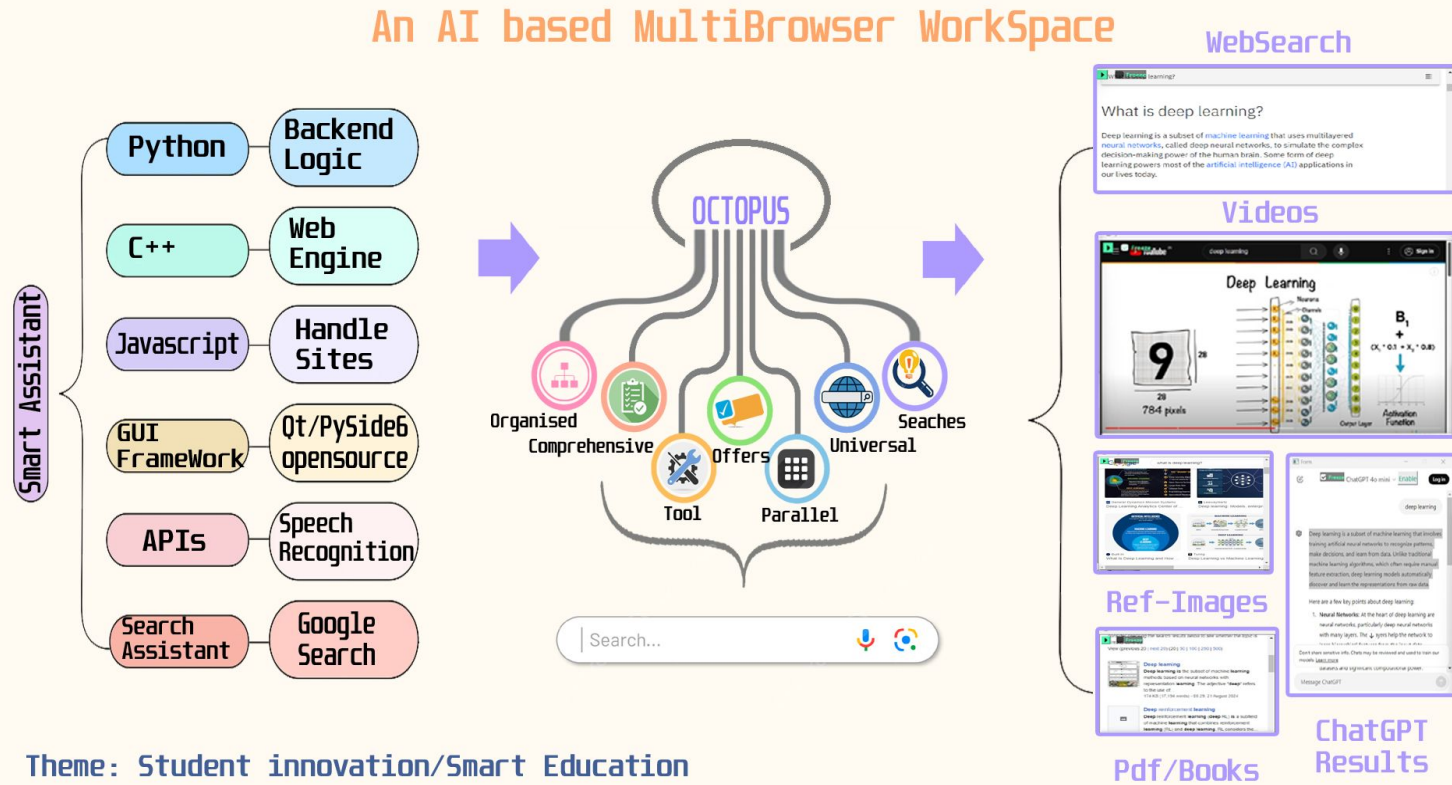
### Programming Languages

- Python, C++ ,Javascript
- PySide for Open Source Development.
- QT Framework for Ui/Ux.
- Speech recognition & Google search Api for Web Scraping, & Tensorflow, Keras for Machine Learning & AI

### Tools

- 1. Visual Studio Code,
- 2. Figma for Ui
- 3. Canva
- 3. Vizcom.ai
- 4. PhotoPea
- 5. Dreamweaver
- 6. QT Design Studio

## Flow Chart :-



### Step 1

Firstly we design a basic prototype with core working idea to implementation the project.

### Step 2

We Use Figma, PS, Concepts to Design GUI. and for UX & components we use Qt Framework. for rapid UI Upgradations.

### Step 3

we use Python for backend logic, database and GUI connection. c++, javascript for webengine integration

### Step 4

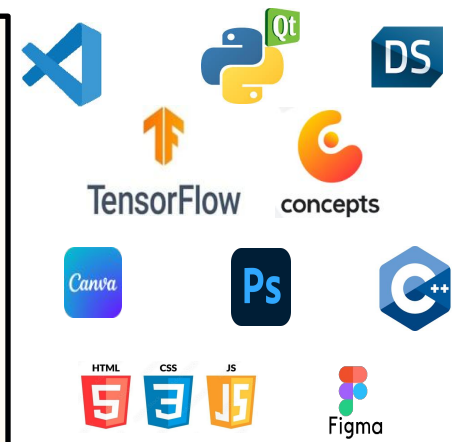
we add some extra features like: Drag & Drop Notes maker, we also add Ai assistant so that it can be operated using Voice commands

### Step 5

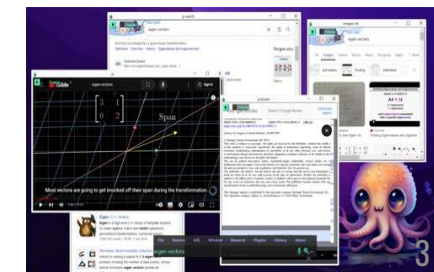
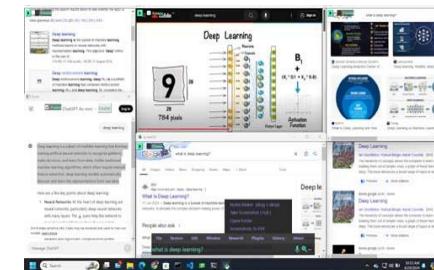
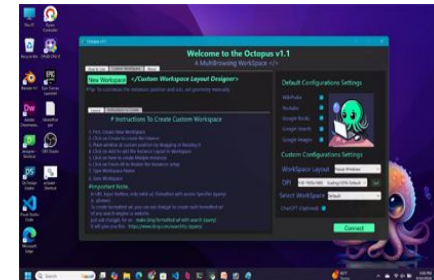
We take feedback from users and experts to regularly upgrade our project features that fits Student needs

### Step 6

At final, we integrate license 7 deploy the project for production environment and distributions.



## Prototype Images:-





# FEASIBILITY AND VIABILITY

## Potential challenges and strategy for overcoming these challenges

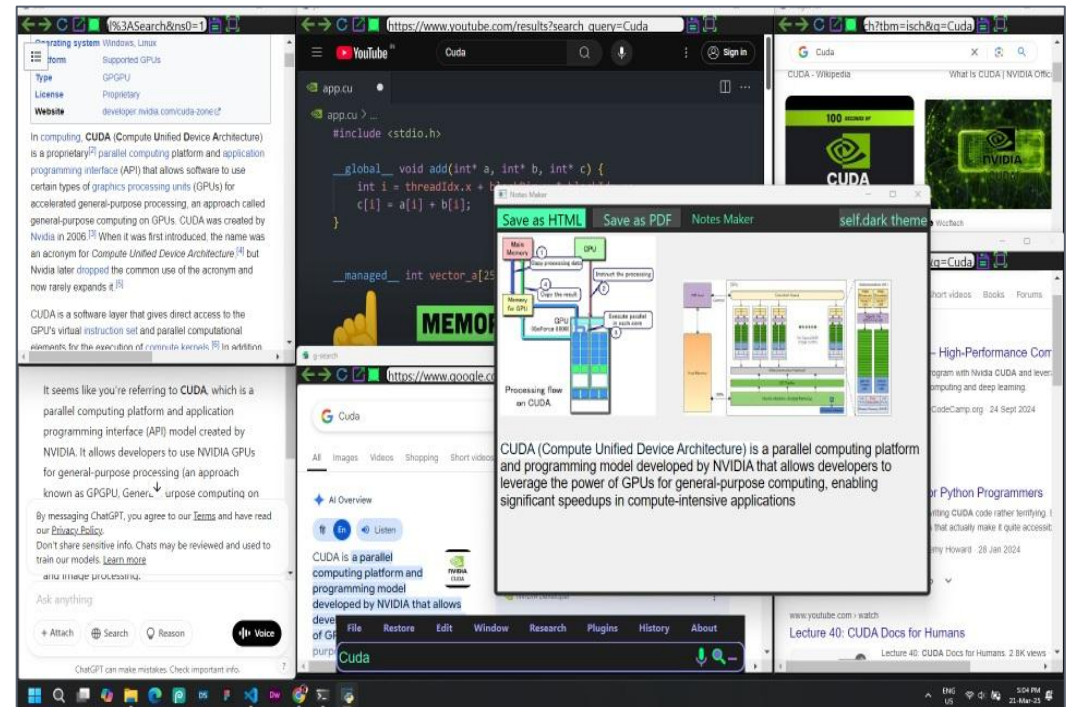
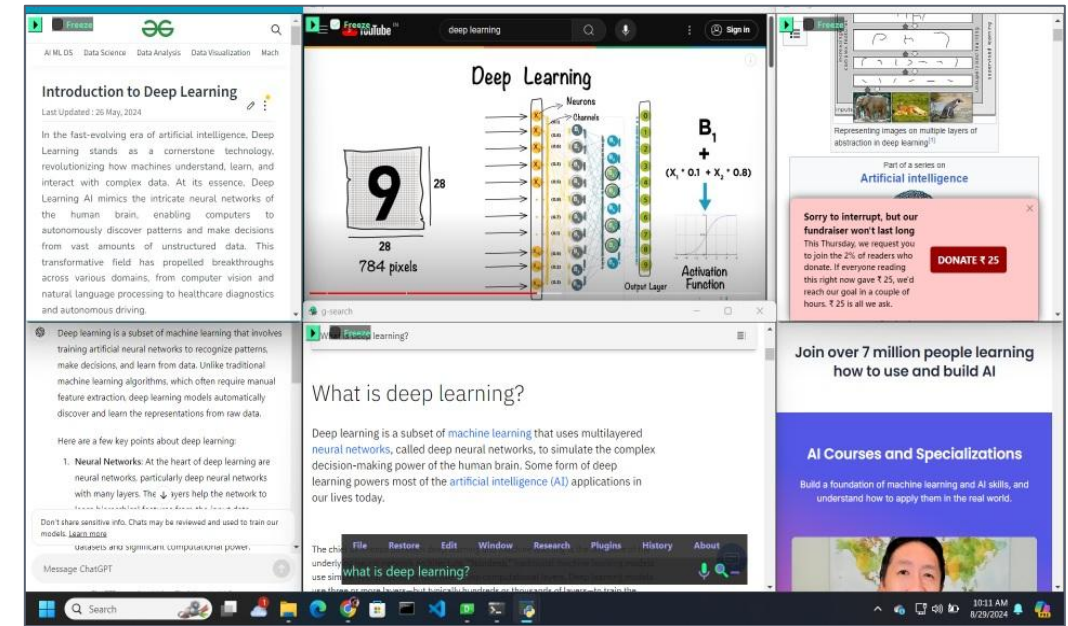
- 1. The 1<sup>st</sup> challenge is to design a web engine and connect multiple Webengines to single Search Bar in such a way that when we Search & Enter, it shows different related information in multiple viewports Instantly.

Other challenges are:

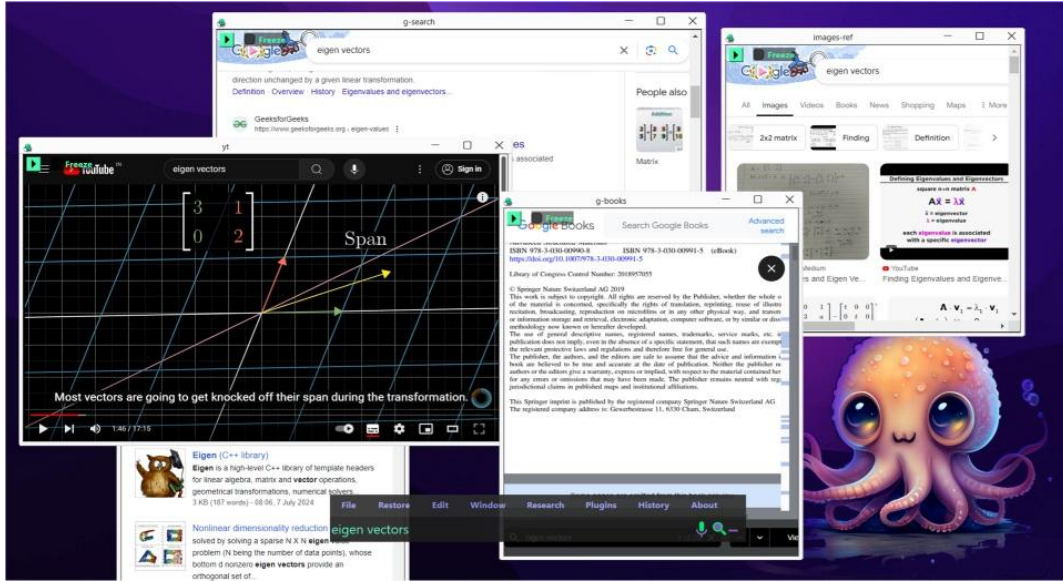
- 2. Performance related, 3. RAM Usage, 4. UX - User Experience

## Some Strategies to overcome these challenges

- For better browsing performance we use C++ based Open Source Web Engine.
- To create backend logic and rapid improvement we use Python. It's also better for upcoming AI Integrations.
- For Better memory management and processing, we use MultiProcessing & Dynamic Programming techniques.
- For GUI and user experience we Use latest Open Source QT Framework.
- We Design our project's UI as Simple and Easy to use.



# IMPACT AND BENEFITS



## Some Future Improvements

- ❑ **Ai Assistant Integration:-** we will integrate Some useful Ai to get instant human like explanations, when asked by user.
- ❑ **Speech Recognition:-** User can ask and talk to the program to perform an action. /Reference: Alexa
- ❑ **Hand Gestures Control:-** User can control the windows using hand sign & gestures. It's very useful when program is projected on a big screen, presenter can control the screen directly by hand gestures. If camera module available at front. /Reference: Apple Vision Pro

## Benefits of the Solution

- ❑ **Comprehensive Results –** Accessing information from various Examples allows for more Comprehensive Understanding of the Topic, and Encourage diverse Perspective.
- ❑ **Time Efficiency-** By Accessing across multiple engines simultaneously, we save our valuable time on Research specific tasks.
- ❑ **Organised For Productivity –** It can be used for various tasks like coding, learning, searching etc. in this, a User can create and design its own workspace layout for a specific type of productive work.
- ❑ **Reliability of Information:-** You will find that information on internet is more reliable when multiple instances or resources shows the similar meaning.
- ❑ **Versatility:-** Whether you are conducting academics research or simply exploring a topic. It adapts to various use cases.
- ❑ **Ease Of Comparison-** Viewing search results Side by Side facilitates quick & Easy comparison, helping in Information Evaluation.



# RESEARCH AND REFERENCES

## Team Abhishek

### Team Members Details:-

- **ABHISHEK VERMA** (Team Leader / Idea / Developer)
- **Atharv Jharbade** (member)
- **Aditya Singh** (member)
- **Akshay Deshmukh** (member)

Institute Name : Oriental Institute of Science & Technology  
Branch : B-Tech Cs-AIML  
Semester : 6<sup>th</sup>  
Minor Project: : Student Innovation theme  
Project Type : Desktop Software Application

Reference : Ironman Movie

Link = <https://youtu.be/rt1dBr2Jz78?si=cXABTtjVoexDyMdB&t=101>

Reference : Apple Vision Pro

Link = <https://makeagif.com/gif/apple-reveals-vision-pro-mixed-reality-headset-Rq6jL2>

### References Images:-



### Prototype Images:-



Prototype Demonstration Link:

<https://youtu.be/nlGrm84790Q>